Eunsong Koh

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Education

Simon Fraser University Expected Graduation Apr. 2027
Bachelor of Science, Computer Science *Vancouver, BC*

- **Relevant Coursework:** Database Systems, Systems Programming, Networking, Computational Data Science, Data Structures & Algorithms
- **Extracurriculars:** SFU DNS Club Dir. Communications & Multimedia, SFU WiCS Mentor & Workshop Lead

Experience

Machine Learning Engineer Intern May 2026 - Jul 2026
Amazon *Vancouver, BC*

- Architected and deployed a **production serverless pipeline** for automated ML-based product classification for tax compliance, reducing cycle time by **90%** and enabling correct tax rate determination for **millions of products** in the Amazon marketplace
- Delivered **two execution modes** (iterative validation and production persistence) sharing a single orchestration layer via **AWS Step Functions** and **Lambda**, deployed across **3 staging environments** with **end-to-end integration tests**
- Built a large-scale **data ingestion pipeline** using **AWS Glue** for monthly catalogue refresh and implemented **polling-based orchestration** with **DynamoDB** correlation tracking to handle asynchronous ML batch inference
- Designed **cross-service integration contracts** across **3 independently-owned systems** with no pre-existing APIs or documentation, coordinating with **4 cross-functional teams** as sole engineer and authoring design documents for each integration point

Applied Research Engineer Jan 2026 - Apr 2026
Safety CLI Cybersecurity | Pre-Series A (Backed by First Round Capital & Kearny Jackson) *Vancouver, BC*

- Redesigned sequential ECS vulnerability analysis into a **distributed graph-processing pipeline** on AWS, improving throughput by **5x** and reducing compute costs by **50%**
- Improved LLM-based vulnerability classification precision from **28% to 78%** through prompt optimization and benchmarking across multiple decoder models; experimented with **BERT-based** fine-tuning and hyperparameter optimization for classification evaluation
- Reduced LLM inference costs by **40%** through model selection and chunking strategy optimizations, while building automated evaluation and experiment tracking pipelines using **LangSmith/LangChain**
- Expanded internal vulnerability triage platform with automated classification and analyst review workflows with **React**, reducing projected manual review workload by **90%+**

Undergraduate Researcher May 2025 - Apr 2026
Reliable Systems Lab | Simon Fraser University *Vancouver, BC*

- Co-authored research on **Agentic-based automated vulnerability discovery and repair** under Dr. Steven Ko, submitted to **NeurIPS 2026** [arXiv]; experimented with **prompt engineering** techniques across multiple strategies to improve model-guided detection by **10%**
- Evaluated open-source and frontier model performance on Rust vulnerability detection using **LangSmith/LangChain** benchmarking pipelines, comparing outputs across multiple metrics to inform model selection

Software Engineer Intern May 2025 - Dec 2025
Safety CLI Cybersecurity | Pre-Series A (Backed by First Round Capital & Kearny Jackson) *Vancouver, BC*

- Developed a **ML-powered data licensing pipeline** with custom parsing algorithms on unstructured data, achieving higher than **90%** evaluation accuracy and generating **multi-million dollar** revenue impact
- Built a **full-stack vulnerability analysis platform** using **FastAPI + React**, leading 90% of frontend development; accelerated security reviews 4× and improved team productivity 16%
- Engineered and productionized a **high-performance call-graph traversal algorithm** for vulnerability detection (50k+ nodes in under 1 second) in the data engine, enabling scalable, real-time codebase analysis
- Prototyped and evaluated decoder and **Hugging Face Transformer** models on unstructured data, applying **distributed training on AWS EC2** with **Accelerate and Ray Train/Tune** to reduce training time by 25%

Software Engineer Intern May 2024 - Aug 2024
Kelowna Software Ltd. *Remote*

- Developed back-end systems in **C#/.NET Core** and client-facing features with **TypeScript**/**Angular** for **1,000+** users across 3 applications
- Automated **ML data pre-processing** using **Python**, **Pandas**, and **Dask**, reducing data sorting and transformation time by **70%**

Projects

AutoJob | *AWS Step Functions, DynamoDB, Bedrock, EventBridge, GitHub Actions*

- Built a **serverless job posting tracking platform** using **AWS Step Functions**, **DynamoDB**, and **Bedrock** for relevance-based classification, **EventBridge** for scheduled notifications, and automated **CI/CD pipelines via GitHub Actions** for reliable large-scale execution

Technical Skills

Languages: C, C++, C#, Python, Java, SQL, JavaScript, TypeScript, Swift, HTML/CSS
Frameworks/Libraries: .NET, Angular, React.js, Next.js, Redux, Django, JUnit/XUnit, Pandas, Bootstrap, TailwindCSS, SwiftUI
Developer Tools/Technologies: AWS, Azure, GCP, LangChain/LangSmith, HuggingFace, MongoDB, Docker, Git, Github Actions, Figma